

H524 GROUP PROJECT: AI COMPARISON IN BIostatISTICS

Fall 2025 — Complete Project Guide

Project Overview

Analyze a medical dataset using statistical methods from Weeks 1-9, while consulting three AI tools (**Claude**, **ChatGPT**, **Microsoft Copilot**). Critically compare their outputs: Which methods did each suggest? Did they agree? Which was most appropriate? Document findings in a written report and oral presentation.

Key Focus: AI comparison. This project tests your ability to critically evaluate AI outputs, not just use them.

Group Assignments & Datasets

Group	Dataset
1	Anorexia Treatment Study
2	Pima Diabetes Study
3	Exercise & Cholesterol Study
4	Vitamin D Supplementation Study

Check Canvas for your group assignment. You'll see your group number and teammates posted.

Timeline & Check-Ins

Week	Activity & Deliverable
Week 7	CHECK-IN: Team Plan (5 pts, part of Week 7 assignment). Leader uploads: team roster, communication plan, meeting schedule, role assignments. Due Sunday 11:59pm.
Week 8	CHECK-IN: Progress Update (1 pt, part of Week 8 assignment). Leader uploads: brief update on progress (has group met? data loaded? AI consulted?). Due Sunday 11:59pm.
Week 9	CHECK-IN: Presentation Readiness (3 pts, part of Week 9 assignment). Leader uploads: presentation outline OR draft slides OR plan. Due Sunday 11:59pm.
Week 10	PRESENTATIONS (10% of course grade): Mon 12-1:20pm (Groups 1&2), Wed 12-1:20pm (Groups 3&4). 15 min + 5 min Q&A. All team members participate. Use slides.
Finals Week	FINAL REPORT (15% of course grade): Due Wednesday 11:59pm. Upload PDF to Canvas (one per team). 5-8 pages + R code appendix.

Note: Check-ins count toward Weekly Assignments (60%), not Group Project (25%). Submit via Canvas assignment uploads.

Dataset Descriptions & Variables

Data Source: All datasets available in the `MedDataSets` R package. See README files in your group folder for loading instructions and more details.

Group 1: Anorexia Treatment Study (n=72 patients)

- **Treat:** Treatment type (Cont=Control, CBT=Cognitive Behavioral Therapy, FT=Family Therapy)
- **Prewt:** Weight before treatment (lbs)
- **Postwt:** Weight after treatment (lbs)

Group 2: Pima Diabetes Study (n=768 women)

- **glu:** Glucose level (mg/dL)
- **bmi:** Body mass index
- **age:** Age (years)

- **type:** Diabetes diagnosis (Yes/No)

Group 3: Exercise & Cholesterol Study (n=50 adults)

- **exercise:** Weekly exercise hours (continuous)
- **cholesterol:** Total cholesterol (mg/dL)
- **age:** Age (years)
- **sex:** Sex (M/F)

Group 4: Vitamin D Supplementation Study (n=45 adults)

- **group:** Supplement dose (Low, Medium, High)
- **vitamin_d:** Vitamin D level after 8 weeks (ng/mL)
- **age:** Age (years)
- **baseline:** Baseline vitamin D level (ng/mL)

AI Tools Overview

1. **Claude** (claude.ai) — Statistical reasoning, assumption checking
2. **ChatGPT** (chat.openai.com) — Code generation, multiple approaches
3. **Microsoft Copilot** (RStudio) — Code completion, inline suggestions

AI Comparison Methodology: Iterative Refinement

Don't just ask once and compare. Use this systematic process:

Step 1: Initial Consultation — Ask Tools 1 & 2 same question. Document both responses.

Step 2: Comparison — Use Tool 3 as “judge.” Prompt: “Tool 1 suggested [X]. Tool 2 suggested [Y]. Compare these. Which is more appropriate? Score each 0-10 and explain.”

Step 3: Refine — If tools disagree, refine prompt to be more specific. Re-ask both tools. Use Tool 3 to compare again.

Step 4: Convergence — Keep refining until: (a) tools give similar advice OR (b) you understand why they differ. Convergence = understanding trade-offs, not identical answers.

Step 5: Document — Save ALL prompts/responses (not just final). Note iterations, what changed, when you converged. This becomes your report's AI Comparison section.

Example: Round 1: Both suggest ANOVA. Round 2: Tool 1 lists normality + equal variance; Tool 2 misses equal variance. Round 3: Tool 3 judges Tool 1 more complete. *Convergence reached: Tool 1 more thorough.*

Suggested First Question (to get started):

“I have uploaded a dataset with [describe your variables]. What are some good research questions I could investigate with this data? For each research question you suggest, what statistical method would be appropriate and what assumptions would I need to check?”

Example for Group 1 (Anorexia): “I have uploaded a dataset with 3 treatment groups (Control, CBT, Family Therapy). Each patient has a pre-treatment weight and post-treatment weight. What are some good research questions I could investigate? For each question, what statistical method should I use and what assumptions need checking?”

This exploratory approach gives you a starting point for your iterative comparison process.

Report Requirements (15% of course grade)

Due: Finals Week Wednesday 11:59pm — **Format:** PDF upload to Canvas

Required Sections:

1. Introduction

Dataset description, research questions

2. Methods

Statistical approaches used, why appropriate, how you consulted AI tools

3. Results

Key findings with tables/figures, test results, visualizations

IMPORTANT: Present your final analysis clearly first (as if for stakeholders), then discuss AI process in Section 4

4. AI Tool Comparison — MOST IMPORTANT

What did each tool suggest? Where did they agree/disagree? Which was most appropriate? Any errors? Which was most helpful? What did this teach you?

Example: “Claude suggested paired t-tests for each treatment group, ChatGPT suggested ANOVA on change scores. Claude’s approach simpler but multiple tests. We chose ANOVA to compare all groups simultaneously and verified assumptions.”

5. Discussion

Interpretation of findings in context, limitations, conclusions

6. Appendix: R Code

Complete, well-commented, reproducible code

Presentation (10% of course grade)

When: Mon Week 10 12-1:20pm (Groups 1&2), Wed Week 10 12-1:20pm (Groups 3&4)

Format: 15 min + 5 min Q&A. All team members participate. Use slides.

Content: Dataset overview (2-3 min), Statistical findings — your final answer to the research question (5-6 min), **AI comparison highlights — what the tools suggested and how you chose (5-6 min, 30-40%)**, Conclusions (1-2 min)

Note: Present your final results clearly first (as if for stakeholders), then show the AI comparison process that got you there.

Tips: Practice timing. Divide speaking roles. Focus on AI comparison. Show examples (screenshots, code comparisons).

Grading

Report (15%): Statistical analysis (5%), AI comparison depth (7%), Writing quality (3%)

Presentation (10%): Content quality (5%), AI comparison focus (3%), Delivery/teamwork (2%)

All team members receive same grade unless participation issues documented.

Getting Help & Tips

Help Available: Week 9 Wed 12-1:20pm drop-in (153 Milam)

Success Tips: Start early. Use all 3 AI tools extensively. Document as you go (save AI chats, note errors). Verify everything — AI makes mistakes. Meet regularly (Weeks 7-10). Focus on AI comparison.

Remember: The goal is not to find the “right” AI tool, but to learn how to **critically evaluate** AI outputs and **verify** their suggestions. All three tools have strengths and weaknesses — your job is to synthesize the best approach from multiple sources and understand when AI helps vs. when it misleads.